TECHNICAL DILIGENCE REVIEW - RECORD 019

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-019

1. Engagement scope, assessors and evidence conventions

The subject of the present report is the security architecture, penetration-test findings and vulnerability management of the platform, examined for Ardent Peak by Pinehurst Systems Assurance over 7 weeks ending 2026-05-08. As instructed, Pinehurst Systems Assurance confined this report to the security architecture, penetration-test findings and vulnerability management of the platform; three adjacent questions raised during the work are listed as follow-ups rather than answered here. One declinature in the present report rests on an engineering judgement the reviewer shares; the others rest on prioritisation, which is a different kind of argument. The report at TDR-2026-19 records 4 items the engineering team declined to remediate, each with the reason given and the reviewer's view of it. L. Casaubon has recorded the date of each measurement, because several of them will be stale within a quarter.

Work on the present report ran over 7 weeks and included 13 interviews and read-only access to the engineering repositories. L. Casaubon was granted read access to the code, the pipeline and the incident tracker for the purposes of the report at TDR-2026-19; production access was not requested. For the transaction, TDR-2026-19 identifies 1 matters the reviewer would raise before closing and 6 for the post-close engineering plan. Deal implications in this report are modest in aggregate and specific in detail: 1 items for the closing checklist, 6 for the plan. L. Casaubon has avoided aggregating findings into a single score, because the aggregate would conceal the distribution.

Readers of TDR-2026-19 should note that a moderate-evidence finding may carry a high severity, and that this is not a contradiction. The present report reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. The pack the report at TDR-2026-19 proposes is deliberately short, because a longer one would be produced and not read. Pinehurst Systems Assurance recommends that the 5 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. Pinehurst Systems Assurance was given read access without restriction on scope within the agreed subject matter.

L. Casaubon chose the sample for this report after reading the dependency map, so the selection is not proportionate to headcount or to code volume. 18 services were examined in detail for the present report, chosen for their position in the dependency map rather than at random. Two positions in this report are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. The reviewer asked in each case whether a declined item had a compensating control, and this report reports the answer even where it is unsatisfactory. L. Casaubon interviewed 13 people for the present report; where an account was uncorroborated it is marked as such.

The principal limitation on report TDR-2026-19 is time: 7 weeks was enough to test the control path and not enough to test the reporting path. Nothing in this report rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. Report TDR-2026-19 added two components to its sample after the first week, because the interview evidence pointed at them. Selection for the present report favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. This report deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

Interviews for the report at TDR-2026-19 were conducted with the owning engineer present rather than with a manager speaking on their behalf. The reviewer found the interview evidence for the present report candid, and in two cases more candid than the written record. 5 indicators are recommended in this report; the reviewer deliberately avoided proposing anything that needs a new pipeline. This report recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. The security architecture, penetration-test findings and vulnerability management of the platform produced more evidence than the present report reproduces, and the working papers hold the remainder.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2026-19)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-19-001	Fleet Allocator: interview minute	interview minute	2025-08-12	Helioscope engineering	evidenced a remediation
EV-19-002	contract tests, warehouse	vendor statement	2025-09-05	Helioscope field reliability	corroborated an incident narrative

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
	connectors: vendor statement				
EV-19-003	Config Distribution: design record	design record	2025-07-18	the sensor vendor	verified a stated objective
EV-19-004	Traffic Manager: runbook	runbook	2026-10-20	Redwater Security	corroborated an incident narrative
EV-19-005	Costmap Fusion: release record	release record	2025-08-03	Helioscope security	supported a capacity conclusion
EV-19-006	static analysis, robot software: runbook	runbook	2025-07-22	Redwater Security	supported a coverage figure
EV-19-007	Billing Usage Meter: design record	design record	2025-05-07	Helioscope safety	evidenced an approval
EV-19-008	Battery Management Service: incident post-mortem	incident post-mortem	2025-11-12	Helioscope security	verified a stated objective
EV-19-009	simulation suite, short: incident post-mortem	incident post-mortem	2025-05-31	the sensor vendor	corroborated an incident narrative
EV-19-010	dependency and licence scan: architecture note	architecture note	2025-08-16	Helioscope field reliability	established a dependency
EV-19-011	Safety Controller: release record	release record	2026-01-21	Helioscope security	established a dependency
EV-19-012	static analysis, robot software: architecture note	architecture note	2025-12-08	Meridian Assurance	established provenance
EV-19-013	hardware-in-the-loop, bench: architecture note	architecture note	2025-08-06	the sensor vendor	supported a capacity conclusion
EV-19-014	Traffic Manager: vendor statement	vendor statement	2025-07-13	Helioscope security	supported a coverage figure
EV-19-015	Video Snippet Service: interview minute	interview minute	2026-08-23	Helioscope safety	established provenance
EV-19-016	Alarm Router: test report	test report	2026-06-17	Helioscope safety	evidenced an approval
EV-19-017	Lidar Driver: release record	release record	2026-07-01	the safety controller vendor	evidenced an approval
EV-19-018	build, robot firmware images: policy document	policy document	2025-08-29	Helioscope security	supported a capacity conclusion
EV-19-019	release packaging and signing: incident post-mortem	incident post-mortem	2025-12-01	the sensor vendor	corroborated an incident narrative
EV-19-020	static analysis, robot software: configuration export	configuration export	2026-03-18	Helioscope safety	supported a capacity conclusion
EV-19-021	Remote Operations Console: vendor statement	vendor statement	2026-06-30	the sensor vendor	established provenance
EV-19-022	Map Authoring Studio: audit output	audit output	2025-06-18	the sensor vendor	established provenance
EV-19-023	Gripper Controller: design record	design record	2025-07-31	Helioscope field reliability	supported a coverage figure
EV-19-024	WMS Connector: policy document	policy document	2026-04-21	Helioscope field reliability	evidenced an approval
EV-19-025	contract tests, warehouse connectors: incident post-mortem	incident post-mortem	2026-10-07	Helioscope security	verified a stated objective
EV-19-026	Wheel Odometry Filter: interview minute	interview minute	2025-06-07	Redwater Security	supported a coverage figure
EV-19-027	Simulation Harness: vendor statement	vendor statement	2026-05-06	the sensor vendor	established provenance
EV-19-028	Speed and Separation Monitor: test report	test report	2026-01-13	Meridian Assurance	established a dependency
EV-19-029	Site Provisioning Service: release	release record	2026-10-31	the sensor vendor	supported a coverage figure

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
	record				
EV-19-030	Safety Controller: interview minute	interview minute	2026-03-10	Helioscope engineering	established a dependency
EV-19-031	simulation suite, short: measurement extract	measurement extract	2026-02-22	Redwater Security	supported a coverage figure
EV-19-032	Safety Controller: configuration export	configuration export	2025-07-20	Meridian Assurance	corroborated an incident narrative
EV-19-033	Robot Shadow Store: architecture note	architecture note	2026-05-18	the sensor vendor	supported a coverage figure
EV-19-034	Simulation Harness: runbook	runbook	2026-08-31	the safety controller vendor	supported a capacity conclusion
EV-19-035	Simulation Harness: architecture note	architecture note	2026-06-29	Redwater Security	supported a capacity conclusion
EV-19-036	Edge Gateway Agent: policy document	policy document	2025-05-29	Helioscope field reliability	established a dependency

1. Engagement scope, assessors and evidence conventions - part 2 of 2

Where the 1. Engagement scope, assessors and evidence conventions was interrupted by the page, it picks up again below, and the numbering is continuous throughout the schedule. The parts are to be read together.

Whatever caveat the 1. Engagement scope, assessors and evidence conventions carried at its head travels with these rows, and no additional caveat attaches to this part. The break is editorial.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2026-19) (continued, part 2 of 2)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-19-037	OTA Updater: incident post-mortem	incident post-mortem	2025-12-05	Helioscope field reliability	established provenance
EV-19-038	Billing Usage Meter: design record	design record	2025-10-19	Meridian Assurance	supported a coverage figure
EV-19-039	Alarm Router: policy document	policy document	2025-12-17	Helioscope field reliability	supported a capacity conclusion
EV-19-040	Identity and Authorization: runbook	runbook	2025-07-22	Meridian Assurance	evidenced an approval
EV-19-041	unit tests, cloud services: design record	design record	2026-11-03	Helioscope security	established provenance
EV-19-042	Wheel Odometry Filter: measurement extract	measurement extract	2025-06-26	Redwater Security	supported a capacity conclusion
EV-19-043	Alarm Router: test report	test report	2025-05-30	Helioscope security	supported a capacity conclusion
EV-19-044	Traffic Manager: configuration export	configuration export	2025-07-15	Helioscope engineering	established provenance
EV-19-045	Analytics Warehouse Loader: release record	release record	2025-09-22	Helioscope engineering	established provenance
EV-19-046	Costmap Fusion: configuration export	configuration export	2026-03-03	Helioscope security	verified a stated objective
EV-19-047	build, cloud container images: vendor statement	vendor statement	2026-01-06	Redwater Security	corroborated an incident narrative
EV-19-048	Gripper Controller: vendor statement	vendor statement	2026-07-02	Helioscope security	established provenance
EV-19-049	Alarm Router: design record	design record	2025-06-11	Redwater Security	evidenced an approval
EV-19-050	Robot Shadow Store: interview minute	interview minute	2026-09-12	Helioscope engineering	evidenced a remediation
EV-19-051	Hardware-in-the-Loop Rig: runbook	runbook	2026-10-06	Helioscope field reliability	established provenance
EV-19-052	Battery Management Service: runbook	runbook	2025-12-18	Helioscope safety	evidenced an approval
EV-19-053	Lidar Driver: measurement extract	measurement extract	2026-04-10	Helioscope safety	supported a capacity conclusion
EV-19-054	dependency and	release record	2026-01-03	Redwater Security	supported a coverage

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
	licence scan: release record				figure
EV-19-055	WMS Connector: audit output	audit output	2026-05-27	Meridian Assurance	evidenced a remediation
EV-19-056	Gripper Controller: configuration export	configuration export	2025-11-12	Helioscope field reliability	established provenance
EV-19-057	Safety Controller: vendor statement	vendor statement	2025-08-01	Helioscope security	evidenced a remediation
EV-19-058	dependency and licence scan: release record	release record	2025-10-14	Helioscope security	supported a coverage figure
EV-19-059	Telemetry Ingest: architecture note	architecture note	2026-07-05	Helioscope field reliability	verified a stated objective
EV-19-060	performance smoke, cloud: audit output	audit output	2025-07-26	Meridian Assurance	established provenance
EV-19-061	Map Authoring Studio: release record	release record	2026-09-10	the safety controller vendor	evidenced a remediation
EV-19-062	Release Orchestrator: incident post-mortem	incident post-mortem	2026-10-26	Helioscope security	corroborated an incident narrative
EV-19-063	WMS Connector: incident post-mortem	incident post-mortem	2025-10-12	Meridian Assurance	supported a coverage figure
EV-19-064	Analytics Warehouse Loader: test report	test report	2025-11-23	Redwater Security	evidenced a remediation
EV-19-065	static analysis, robot software: design record	design record	2025-06-15	Helioscope field reliability	established provenance
EV-19-066	lint and format: interview minute	interview minute	2026-08-30	Meridian Assurance	supported a coverage figure
EV-19-067	OTA Updater: interview minute	interview minute	2026-01-06	Helioscope security	established provenance
EV-19-068	build, cloud container images: measurement extract	measurement extract	2026-09-05	the sensor vendor	evidenced a remediation
EV-19-069	simulation suite, nightly: policy document	policy document	2026-08-24	the safety controller vendor	supported a coverage figure
EV-19-070	OTA Updater: measurement extract	measurement extract	2025-08-05	the sensor vendor	verified a stated objective
EV-19-071	contract tests, warehouse connectors: incident post-mortem	incident post-mortem	2026-10-22	Helioscope engineering	corroborated an incident narrative
EV-19-072	Battery Management Service: incident post-mortem	incident post-mortem	2026-06-04	Redwater Security	supported a capacity conclusion

2. SEC-2026-07 and the high-severity findings

Finding SEC-2026-07 is rated Medium with a numeric band of 6.7 on the tester's scoring scale. SEC-2026-07 against Hardware-in-the-Loop Rig rests on the observation that operating system version observed on the rig; the reviewer holds that to be moderate. Hardware-in-the-loop rig controller runs an unsupported operating system is the substance of finding SEC-2026-07, raised on 2026-04-17. Tracking Medium-severity finding SEC-2026-07 to closure gives a status of open and a date of committed 2026-10-31. Hardware-in-the-Loop Rig finding SEC-2026-07 has no single accountable owner, and that is the finding the reviewer would press hardest.

After the fix, finding SEC-2025-03 carries residual risk described as low; pinning is applied at the gateway and the agent. The practical consequence in Remote Operations Console finding SEC-2025-03 is that an on-path attacker on a customer network could observe operator commands. The remediation ledger shows Medium-severity finding SEC-2025-03 as remediated, 2026-01-22. Weighing finding SEC-2025-03 against the compensating controls, management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. Remote Operations Console finding SEC-2025-03 is treated as material to the investment case rather than as a routine engineering item.

What remains after SEC-2026-13 is remediated is low. The reviewer declines to soften SEC-2026-13 against Order Broker: the item is open past its committed date and no revised date has been agreed. SEC-2026-13 against Order Broker arrives as a Low finding with a 3.3 score, and the reviewer did not adjust either. Substantiating Low-severity finding

SEC-2026-13 is the fact that throughput degradation measured by the tester, graded strong. finding SEC-2026-13 is the reason one section of TDR-2026-19 concludes narrowly rather than generally.

Reading assessment finding SEC-2026-11, the defect is that gripper controller parameter store writable by any on-robot process; the discovery date is 2026-04-19. Gripper Controller finding SEC-2026-11 is supported by the fact that file permissions observed on a laboratory robot; evidence strength is strong. On Low-severity finding SEC-2026-11 the conclusion is that the underlying pattern appears in two other components that the tester did not reach. The practical consequence in finding SEC-2026-11 is that a non-privileged process could alter a force limit parameter. assessment finding SEC-2026-11 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Remediated is the status of Medium-severity finding SEC-2026-08, dated 2026-06-30. On SEC-2026-08 the conclusion is that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. Medium-severity finding SEC-2026-08 is rated Medium with a numeric band of 6.4 on the tester's scoring scale. The reason Medium-severity finding SEC-2026-08 was not downgraded is that one administrator can grant themselves tenant-wide access. Medium-severity finding SEC-2026-08 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Assessment finding SEC-2025-07 was graded Medium, band 5.9, by Redwater Security engagement RW-2025-3. The residual position on Medium-severity finding SEC-2025-07 is medium; two forked templates were not re-issued until May 2026, stated after mitigation rather than assuming full success. On 2025-09-22 the tester recorded SEC-2025-07 against Site Provisioning Service: site provisioning stored a shared bootstrap secret in the template. SEC-2025-07 against Site Provisioning Service matters because the same secret was reusable at any site provisioned from that template. SEC-2025-07 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Rating for Medium-severity finding SEC-2025-09 is Medium, band 6.3; the reviewer re-read the scoring rationale and accepted it. Moderate evidence supports finding SEC-2025-09, namely that key custody was evidenced by interview and the key ceremony record. The residual position on SEC-2025-09 is medium; a two-person rule is designed but not enforced in the tool, stated after mitigation rather than assuming full success. Medium-severity finding SEC-2025-09 is partially remediated, with a closure date of pending. SEC-2025-09 against OTA Updater is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

As at the review date Low-severity finding SEC-2026-18 was open and the closure entry reads committed 2026-12-15. SEC-2026-18 reduces to a residual risk of low; no fleet impact demonstrated. Low-severity finding SEC-2026-18 is rated Low with a numeric band of 3.4 on the tester's scoring scale. The practical consequence in Battery Management Service finding SEC-2026-18 is that a diagnostic flood can slow pack telemetry reporting. assessment finding SEC-2026-18 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Severity on Medium-severity finding SEC-2026-16 is Medium, scored 5.8 by the assessor. Impact of SEC-2026-16 is not theoretical; on-network access to a live perception diagnostic stream. SEC-2026-16 against Costmap Fusion reduces to a residual risk of low; the socket binds to the loopback interface only. Assessment finding SEC-2026-16 stands remediated; 2026-06-26 is what the record gives for closure. assessment finding SEC-2026-16 is included because a reader of TDR-2026-19 should be able to disagree with the inference without repeating the work.

SEC-2026-14 against Traffic Manager is remediated, with a closure date of 2026-06-18. The reviewer's position on assessment finding SEC-2026-14 is that the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. SEC-2026-14 records that traffic manager admin action lacked an audit record, discovered on 2026-04-21. In SEC-2026-14 against Traffic Manager the exploitation path means a manual corridor override cannot be attributed after the fact. Low-severity finding SEC-2026-14 illustrates why TDR-2026-19 separates the severity of a finding from the confidence behind it.

Schedule K - Penetration-test findings by severity with residual risk (TDR-2026-19)

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
SEC-2025-01	High	7.4	Identity and Authorization	2025-09-16	2025-11-14	low once lifetimes are cut to one hour with refresh

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
SEC-2025-02	High	8.1	Release Orchestrator	2025-09-17	2025-09-19	low; redaction is enforced in the logging library
SEC-2025-03	Medium	6.2	Remote Operations Console	2025-09-18	2026-01-22	low; pinning is applied at the gateway and the agent
SEC-2025-04	Medium	5.4	Map Authoring Studio	2025-09-18	2025-11-27	medium; the shortened publication path is still outstanding
SEC-2025-05	High	7.8	Telemetry Ingest	2025-09-19	2026-02-06	low; attestation is now required on the ingest path
SEC-2025-06	Medium	6.5	Video Snippet Service	2025-09-19	2025-12-18	low; the index is scoped per tenant and the identifiers are opaque
SEC-2025-07	Medium	5.9	Site Provisioning Service	2025-09-22	2026-01-16	medium; two forked templates were not re-issued until May 2026
SEC-2025-08	Medium	6.0	Event Bus	2025-09-23	2025-10-31	low; the interface is restricted to the operations subnet
SEC-2025-09	Medium	6.3	OTA Updater	2025-09-24	pending	medium; a two-person rule is designed but not enforced in the tool
SEC-2025-10	Low	3.7	Charge Scheduler	2025-09-24	2025-10-09	low
SEC-2025-11	Low	3.1	Remote Operations Console	2025-09-25	2025-10-30	low
SEC-2025-12	Low	4.2	Analytics Warehouse Loader	2025-09-25	not planned	medium; management accepted the risk with a compensating audit trail
SEC-2026-01	High	7.1	Release Orchestrator	2026-04-14	committed 2026-08-31	high until the second-approver control lands
SEC-2026-02	High	7.6	Robot Shadow Store	2026-04-15	2026-05-29	low; authorization is checked on both the cache and durable paths
SEC-2026-03	High	8.4	Config Distribution	2026-04-15	2026-06-12	low; the flag was removed and the agent refuses unsigned bundles
SEC-2026-04	Medium	5.6	WMS Connector	2026-04-16	2026-05-20	low; payload logging requires an explicit per-tenant approval
SEC-2026-05	Medium	5.2	Alarm Router	2026-04-16	committed 2026-09-30	medium; per-region credentials are scheduled but not issued
SEC-2026-06	Medium	6.1	Simulation Harness	2026-04-17	pending	medium; the role was scoped down but still reads one production bucket
SEC-2026-07	Medium	6.7	Hardware-in-the-Loop Rig	2026-04-17	committed 2026-10-31	medium; network segmentation is the only current control
SEC-2026-08	Medium	6.4	Identity and Authorization	2026-04-18	2026-06-30	low; role grants now require a second approver and are alerted

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
SEC-2026-09	Low	3.9	Fleet Health Monitor	2026-04-18	2026-05-08	low
SEC-2026-10	Low	4.0	Billing Usage Meter	2026-04-19	2026-06-05	low; object lock is applied and verified nightly
SEC-2026-11	Low	4.4	Gripper Controller	2026-04-19	committed 2026-11-30	medium; the safety controller limit is the compensating control
SEC-2026-12	Low	3.5	Edge Gateway Agent	2026-04-20	pending	medium; a wipe procedure exists but was not applied retroactively
SEC-2026-13	Low	3.3	Order Broker	2026-04-20	2026-05-15	low
SEC-2026-14	Low	3.8	Traffic Manager	2026-04-21	2026-06-18	low
SEC-2026-15	Low	3.0	Site Provisioning Service	2026-04-21	not planned	low; management judged the operational cost of enforcement higher than the risk
SEC-2026-16	Medium	5.8	Costmap Fusion	2026-04-22	2026-06-26	low; the socket binds to the loopback interface only
SEC-2026-17	High	7.9	Motion Planner	2026-04-22	2026-06-19	low; the port requires a signed local credential
SEC-2026-18	Low	3.4	Battery Management Service	2026-04-23	committed 2026-12-15	low; no fleet impact demonstrated
SEC-2026-19	Medium	5.1	Localization Engine	2026-04-23	committed 2026-10-15	medium; disk encryption is planned for the next compute revision
SEC-2026-20	Low	3.2	Payload Scale Interface	2026-04-24	not planned	low; management accepted pending the calibration service redesign

3. Medium and low findings, ageing and residual risk

Moderate evidence supports finding SEC-2026-15, namely that collision created in a staging tenant. Rating for SEC-2026-15 is Low, band 3.0; the reviewer re-read the scoring rationale and accepted it. On 2026-04-21 the tester recorded SEC-2026-15 against Site Provisioning Service: site provisioning API allowed a site name collision across tenants. Low-severity finding SEC-2026-15 leaves residual risk that the reviewer assesses as low; management judged the operational cost of enforcement higher than the risk. Low-severity finding SEC-2026-15 is the clearest instance in TDR-2026-19 of a control that exists on paper and is weaker in practice.

Rating for Telemetry Ingest finding SEC-2025-05 is High, band 7.8; the reviewer re-read the scoring rationale and accepted it. SEC-2025-05 against Telemetry Ingest is supported by the fact that the tester submitted synthetic telemetry from an unenrolled host; evidence strength is limited, resting on a configuration read rather than on a demonstrated exploitation. On 2025-09-19 the tester recorded finding SEC-2025-05: telemetry ingest accepted a robot identity without checking device attestation. The practical consequence in Telemetry Ingest finding SEC-2025-05 is that a forged agent could inject telemetry attributed to a real robot. High-severity finding SEC-2025-05 was cross-checked against the engineering record supplied for TDR-2026-19, and the two agree.

Emergency release path permits a single approver is the substance of High-severity finding SEC-2026-01, raised on 2026-04-14. Limited, resting on a configuration read rather than on a demonstrated exploitation evidence supports SEC-2026-01 against Release Orchestrator, namely that the tester used the path in a staging tenant and the audit trail recorded no justification. Remediation status on SEC-2026-01 is open; the record shows committed 2026-08-31. High is

the rating SEC-2026-01 carries, at 7.1 on the assessor's own scale. assessment finding SEC-2026-01 is the clearest instance in TDR-2026-19 of a control that exists on paper and is weaker in practice.

The reviewer's position on Localization Engine finding SEC-2026-19 is that the item is open past its committed date and no revised date has been agreed. Evidence behind SEC-2026-19 is that cache read from a removed module, which the reviewer treats as limited, resting on a configuration read rather than on a demonstrated exploitation. SEC-2026-19 arrives as a Medium finding with a 5.1 score, and the reviewer did not adjust either. Committed 2026-10-15 is the closure position on Medium-severity finding SEC-2026-19, which the ledger records as open. Medium-severity finding SEC-2026-19 was checked against the corresponding entry in the company's own register, and the register was current.

High is the rating Config Distribution finding SEC-2026-03 carries, at 8.4 on the assessor's own scale. The reviewer declines to soften Config Distribution finding SEC-2026-03: the fix is verified and the class of defect is not, since no test now prevents its return. A local compromise on a robot becomes a persistent configuration compromise is the exposure SEC-2026-03 against Config Distribution describes. After the fix, finding SEC-2026-03 carries residual risk described as low; the flag was removed and the agent refuses unsigned bundles. assessment finding SEC-2026-03 is reported here in full because a summary of it in an earlier draft of TDR-2026-19 was misread as an assurance.

SEC-2025-06 against Video Snippet Service is assessed as follows: the item is open past its committed date and no revised date has been agreed. Assessment finding SEC-2025-06 reduces to a residual risk of low; the index is scoped per tenant and the identifiers are opaque. 2025-12-18 is the closure position on Medium-severity finding SEC-2025-06, which the ledger records as remediated. Video Snippet Service finding SEC-2025-06 records that video snippet index allowed cross-tenant enumeration by identifier, discovered on 2025-09-19. SEC-2025-06 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Finding SEC-2026-05 stands open; committed 2026-09-30 is what the record gives for closure. For SEC-2026-05 against Alarm Router the record shows that credential reuse confirmed from the secret store inventory, which is strong evidence and is graded separately from severity. Assessment finding SEC-2026-05 is rated Medium with a numeric band of 5.2 on the tester's scoring scale. The residual position on Medium-severity finding SEC-2026-05 is medium; per-region credentials are scheduled but not issued, stated after mitigation rather than assuming full success. finding SEC-2026-05 is treated as material to the investment case rather than as a routine engineering item.

On 2025-09-25 the tester recorded assessment finding SEC-2025-11: console session cookie lacked a strict same-site attribute. Impact of assessment finding SEC-2025-11 is not theoretical; a cross-site request could reuse an active operator session. Substantiating SEC-2025-11 against Remote Operations Console is the fact that attribute absence confirmed; no working exploit chain was demonstrated, graded limited, resting on a configuration read rather than on a demonstrated exploitation. Low-severity finding SEC-2025-11 is remediated, with a closure date of 2025-10-30. Low-severity finding SEC-2025-11 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Assessment finding SEC-2026-04 is supported by the fact that payloads recovered from a staging log store; evidence strength is moderate. On 2026-04-16 the tester recorded SEC-2026-04 against WMS Connector: wMS connector logged customer order payloads at debug level. Customer commercial data is retained in operational logs is the exposure SEC-2026-04 describes. For WMS Connector finding SEC-2026-04 the reviewer notes that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. SEC-2026-04 is the clearest instance in TDR-2026-19 of a control that exists on paper and is weaker in practice.

2026-04-20 is the discovery date of SEC-2026-12, which finds that edge gateway agent retained mutual authentication keys after decommissioning. Partially remediated is the status of assessment finding SEC-2026-12, dated pending. SEC-2026-12 against Edge Gateway Agent arrives as a Low finding with a 3.5 score, and the reviewer did not adjust either. The evidence for Low-severity finding SEC-2026-12 is moderate: keys recovered from a returned module. finding SEC-2026-12 appears twice in the working papers for TDR-2026-19 because the first answer was corrected after a follow-up request.

Schedule L - Vulnerability ageing against the internal remediation window (TDR-2026-19)

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-01	High	22	30 days	yes	manual review before release
SEC-2025-02	High	142	30 days	no	audit trail only
SEC-2025-03	Medium	87	90 days	yes	network segmentation
SEC-2025-04	Medium	43	90 days	yes	restricted role membership
SEC-2025-05	High	76	30 days	no	restricted role membership
SEC-2025-06	Medium	22	90 days	yes	monitoring alert on use
SEC-2025-07	Medium	24	90 days	yes	second delivery path
SEC-2025-08	Medium	154	90 days	no	monitoring alert on use
SEC-2025-09	Medium	308	90 days	no	none recorded
SEC-2025-10	Low	167	180 days	yes	restricted role membership
SEC-2025-11	Low	78	180 days	yes	none recorded
SEC-2025-12	Low	354	180 days	no	restricted role membership
SEC-2026-01	High	232	30 days	no	second delivery path
SEC-2026-02	High	127	30 days	no	none recorded
SEC-2026-03	High	27	30 days	yes	second delivery path
SEC-2026-04	Medium	135	90 days	no	network segmentation
SEC-2026-05	Medium	100	90 days	no	restricted role membership
SEC-2026-06	Medium	192	90 days	no	none recorded
SEC-2026-07	Medium	158	90 days	no	none recorded
SEC-2026-08	Medium	151	90 days	no	manual review before release
SEC-2026-09	Low	150	180 days	yes	audit trail only
SEC-2026-10	Low	78	180 days	yes	network segmentation
SEC-2026-11	Low	200	180 days	no	none recorded
SEC-2026-12	Low	252	180 days	no	second delivery path
SEC-2026-13	Low	24	180 days	yes	network segmentation
SEC-2026-14	Low	128	180 days	yes	monitoring alert on use
SEC-2026-15	Low	127	180 days	no	restricted role membership
SEC-2026-16	Medium	148	90 days	no	network segmentation
SEC-2026-17	High	147	30 days	no	none recorded
SEC-2026-18	Low	305	180 days	no	monitoring alert on use
SEC-2026-19	Medium	116	90 days	no	manual review before release
SEC-2026-20	Low	97	180 days	no	none recorded

4. Authentication, authorization and the trust boundaries

Fleet Allocator was assessed against the objective and the defect record, and the reviewer accepts the engineering position while noting that the verifying test does not exist. For Fleet Allocator the supportability position is none identified in the reviewed manifests. Documents and data inspected for Fleet Allocator were the allocator service contract, two design reviews and a week of assignment audit logs. Fleet Allocator is deployed cloud, one regional deployment serving all sites in that region. Inputs to Fleet Allocator come from the order broker and the robot shadow store, and its consumers are the traffic manager, the charge scheduler and the fleet health monitor. Fleet Allocator was excluded from an earlier draft of TDR-2026-19 and reinstated after the reviewer obtained the supporting record.

The Fleet Services team carries Fleet Allocator on its service inventory, and Marta Kellerman answered for it in interview. Fleet Services releases Fleet Allocator weekly, behind a change approval, and no release in the sampled window bypassed the gate set. The objective recorded against Fleet Allocator is that task assignment inside 900 ms of

order arrival at p95, and it is reported weekly to the engineering leadership. Fleet Allocator reports 82 per cent line coverage, measured by the same instrumentation the pipeline uses. Fleet Allocator pairs a Go 1.22 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in PostgreSQL 14 with a synchronous standby. Fleet Allocator was one of two items where TDR-2026-19 could not obtain the underlying data within the engagement.

Reading the dependency map from Remote Operations Console outward gives the identity service, the telemetry ingest path and the video snippet service on the input side and the traffic manager and the alarm router on the output side. Remote Operations Console was built in TypeScript with a React front end and has not been rewritten; it still runs on cloud, a static bundle plus a session gateway against PostgreSQL 14 for session records and object storage for clips. Remote Operations Console was assessed against the objective and the defect record, and the reviewer accepts the engineering position while noting that the verifying test does not exist. Remote Operations Console was checked for shared-database coupling, and the ownership boundary held. The Applications team carries Remote Operations Console on its service inventory, and Anjali Sekhon answered for it in interview. Remote Operations Console was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Remote Operations Console sits at 52 per cent line coverage; the trend across the reviewed period was down by two points after a large refactor. Release cadence for Remote Operations Console is twice a week, and the release records bear that out. Remote Operations Console carries a single measurable objective, namely that operator input reaches the robot inside 350 ms at p95 on a supported network. Against Remote Operations Console the backlog runs to 27 items, and it includes a known behaviour whereby the session list keeps rendering a robot that has already left the site until a manual refresh. Evidence examined for Remote Operations Console comprised the console interaction specification, session latency measurements and an accessibility walk-through. Remote Operations Console is included because a reader of TDR-2026-19 should be able to disagree with the inference without repeating the work.

Integrations tracks 23 open defects on WMS Connector, including the acknowledged case where duplicate order suppression relies on a customer-supplied identifier that two tenants reuse. Hosting for WMS Connector is cloud, containers on a managed Kubernetes cluster, and its Java 21 code writes through to PostgreSQL 14 plus a message outbox table. Against WMS Connector, the coverage figure is 69 per cent line coverage, which the figure is respectable but excludes the fallback path, which is where the known defect sits. The Integrations team carries WMS Connector on its service inventory, and Tomas Villareal answered for it in interview. WMS Connector rests on six tenant integration specifications and a month of connector error logs so far as this review is concerned. WMS Connector was tested rather than accepted, and the test is described in the working papers behind TDR-2026-19.

The objective recorded against WMS Connector is that order acknowledgement returned to the warehouse system inside 1.5 s at p95, and it is reported weekly to the engineering leadership. WMS Connector was assessed against the objective and the defect record, and management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. The ageing position for WMS Connector is one customer integration still runs on a connector build from 2024, and the reviewer checked it against the vendor statements on file. WMS Connector runs cloud, one deployment per customer tenant; that placement is what determines whether a fault is one robot or one region. The interface WMS Connector publishes is versioned, and the order broker and the billing usage meter were checked for use of any undocumented field. WMS Connector carries a follow-up request that was satisfied after the draft of TDR-2026-19 was circulated.

Telemetry Ingest is owned by the Data and Telemetry team, with Farida Cisse named as the engineering owner of record. Hosting for Telemetry Ingest is cloud, containers plus a stream processor, and its Go 1.22 with a Rust decoder code writes through to a time-series store with ninety-day retention and object storage beyond that. The dependency map places the edge gateway agent and the event bus above Telemetry Ingest and the fleet health monitor, the analytics warehouse loader and the alarm router below it. Read against its objective, Telemetry Ingest the component is adequate today and carries a supportability liability that is not on the funded plan. Changes to Telemetry Ingest reach production weekly, with schema changes gated on a compatibility check. Telemetry Ingest was excluded from an earlier draft of TDR-2026-19 and reinstated after the reviewer obtained the supporting record.

Telemetry Ingest was checked for shared-database coupling, and the ownership boundary held. Telemetry Ingest rests on the telemetry schema registry, retention configuration and a forced-offline buffer test so far as this review is concerned. Telemetry Ingest sits at 61 per cent line coverage; the trend across the reviewed period was up by three points.

Telemetry Ingest runs cloud, one ingest endpoint per region with an edge buffer on the robot; that placement is what determines whether a fault is one robot or one region. Against Telemetry Ingest the backlog runs to 19 items, and it includes a known behaviour whereby the edge buffer drops the oldest samples after four hours offline without raising an alert. Telemetry Ingest has no single accountable owner, and that is the finding the reviewer would press hardest.

What the reviewer had for Video Snippet Service was the retention configuration, a lifecycle audit across regions and the access log sample; the conclusions below do not reach past it. The ageing position for Video Snippet Service is a transcoder build with known but low-severity upstream advisories, and the reviewer checked it against the vendor statements on file. Taking the evidence together, Video Snippet Service the reviewer accepts the engineering position while noting that the verifying test does not exist. Reading the dependency map from Video Snippet Service outward gives the on-robot camera buffer and the identity service on the input side and the remote operations console and the incident review workflow on the output side. Test coverage on Video Snippet Service is 43 per cent line coverage and the measurement basis was confirmed with Anjali Sekhon. Video Snippet Service was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Implementation of Video Snippet Service rests on Go 1.22 with a media pipeline; the process runs on cloud, containers plus a media transcoder and persists to object storage with a thirty-day lifecycle and a PostgreSQL 14 index. Video Snippet Service releases monthly. Video Snippet Service commits that a requested snippet is available to the operator inside 45 s, and the objective is measured continuously rather than sampled. Applications holds Video Snippet Service, and the ownership entry naming Anjali Sekhon matches the on-call rotation. In production Video Snippet Service is cloud, regional, with an on-robot capture buffer, unchanged since the last architecture review. Video Snippet Service was checked against the corresponding entry in the company's own register, and the register was current.

The reviewer's judgement on Billing Usage Meter is that the component is adequate today and carries a supportability liability that is not on the funded plan. Inputs to Billing Usage Meter come from the order broker and the WMS connector, and its consumers are the finance export and the customer usage report. For Billing Usage Meter the reviewer worked from the metering specification, three monthly reconciliations and the ledger immutability check. Data ownership is clear for Billing Usage Meter: it writes PostgreSQL 14 with an immutable usage ledger and no other service writes there. Billing Usage Meter has a single named owner, Nadia Semenova of Data and Telemetry, which is not true of every service reviewed. Billing Usage Meter is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Billing Usage Meter ships monthly, aligned to the billing cycle; the reviewer sampled the last twelve releases. Billing Usage Meter shows 79 per cent line coverage in the pipeline, and that is below what this component's failure mode warrants. Billing Usage Meter pairs a Go 1.22 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in PostgreSQL 14 with an immutable usage ledger. Of the 7 open items against Billing Usage Meter, the reviewer treats one as material: a re-run of a failed metering window can double-count if the operator skips the reset step. Deployment of Billing Usage Meter is cloud, regional, which sets the blast radius of any defect in it. Billing Usage Meter is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Schedule B - Deployment topology and failure domain by service (TDR-2026-19)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Motion Planner	edge, one instance per robot with no cloud fallback	every three weeks, promoted ring by ring	6	one robot	Priya Raghunathan
Costmap Fusion	edge, co-scheduled with the planner on the compute module	monthly, batched with perception model updates	5	one robot	Daniel Okafor
Fleet Allocator	cloud, one regional deployment serving all sites in that region	weekly, behind a change approval	6	one region	Marta Kellerman
Traffic Manager	cloud, regional, with a degraded on-robot yield behaviour if the link drops	weekly, paired with the allocator release	2	one region	Owen Brantley
Charge Scheduler	cloud, regional, with a conservative local fallback policy on the	fortnightly, without a formal approval gate until March 2026	6	one region	Nadia Semenova

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
	robot				
Safety Controller	edge, dual-channel, with cross-checking between channels	twice a year, with full revalidation of the safety case	1	one robot	Bettina Vogel
E-Stop Chain	edge, hardwired, independent of the compute module	changed only by an engineering change order	6	one robot	Theo Bassett
Lidar Driver	edge, one process per sensor	as needed, historically twice a year	5	one robot	Yuki Harada
Localization Engine	edge, with map distribution from the cloud	every three weeks with the planner	1	one robot	Rahul Menon
WMS Connector	cloud, one deployment per customer tenant	every two weeks, per tenant on request	3	one region	Tomas Villareal
Order Broker	cloud, regional	weekly	3	one region	Grace Lindqvist
Telemetry Ingest	cloud, one ingest endpoint per region with an edge buffer on the robot	weekly, with schema changes gated on a compatibility check	1	one robot	Farida Cisse
OTA Updater	split edge and cloud, with staged rings and a per-site pause control	every three weeks, following the autonomy release	5	one robot	Erik Palmgren
Identity and Authorization	cloud, global control plane with regional caches	weekly, with a security review on every change	4	one region	Sonia Krall
Remote Operations Console	cloud, with a thin session relay in each region	twice a week	2	one region	Anjali Sekhon
Simulation Harness	cloud batch, with a local mode for developers	continuous, from the main branch	4	one region	Colin Achebe
Hardware-in-the-Loop Rig	laboratory, two rigs, one of which lags the current sensor configuration	the harness changes with each firmware release	3	one region	Nathan Oduya

4. Authentication, authorization and the trust boundaries - part 2 of 2

Below, the 4. Authentication, authorization and the trust boundaries continues; nothing separates this part from the last except the page on which it happens to fall. The parts are to be read together.

No basis of compilation changes at this break; whatever the 4. Authentication, authorization and the trust boundaries recorded about scope and method at its head continues to apply.

Schedule B - Deployment topology and failure domain by service (TDR-2026-19) (continued, part 2 of 2)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Fleet Health Monitor	cloud, regional	fortnightly	1	one region	Wanda Njoku
Battery Management Service	split edge and cloud	pack firmware twice a year, the cloud service monthly	4	one robot	Peter Halvorsen
Gripper Controller	edge, only on the two robot variants fitted with a manipulator	quarterly	5	one robot	Ines Duarte
Map Authoring Studio	cloud, single global deployment	weekly	1	one region	Lucia Ferrante
Site Provisioning Service	cloud, global	fortnightly	1	one region	Marcus Duplessis
Event Bus	cloud, one cluster per region, three availability zones	upgraded twice a year	5	one region	Erik Palmgren
Config Distribution	split edge and cloud, with a signed bundle format	weekly	1	one robot	Sam Oyelaran
Robot Shadow Store	cloud, regional,	weekly	6	one region	Marta Kellerman

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
	sharded by site				
Analytics Warehouse Loader	cloud, one loader per region writing to a shared warehouse	weekly	2	one region	Farida Cisse
Billing Usage Meter	cloud, regional	monthly, aligned to the billing cycle	4	one region	Nadia Semenova
Alarm Router	cloud, regional, with a second delivery provider	fortnightly	1	one region	Wanda Njoku
Video Snippet Service	cloud, regional, with an on-robot capture buffer	monthly	6	one region	Anjali Sekhon
Speed and Separation Monitor	edge, cross-checked against the safety controller's own inputs	twice a year, with revalidation	5	one robot	Bettina Vogel
Wheel Odometry Filter	edge	with the firmware, twice a year	6	one robot	Yuki Harada
Payload Scale Interface	edge, only on robots fitted with the weighing deck	as needed	6	one robot	Ines Duarte
Edge Gateway Agent	edge, one per robot, with mutual authentication to the cloud	every three weeks	3	one robot	Sam Oyelaran
Release Orchestrator	cloud, single global deployment	continuous, self-hosting	2	one region	Colin Achebe

5. Supply chain and dependency scanning results

In Event Bus the component kafka-client is used event bus producer and consumer clients, which is the fact the licence analysis rests on. Kafka-client also carries a build-time dependency whose absence would stop the pipeline rather than the product, which is a supportability question rather than a licence one. For kafka-client the obligation is to publish the modified files it distributes, and the reviewer tested whether it is met. kafka-client is the entry that changed most between the first and second document requests behind TDR-2026-19.

OSS-006 records chartkit 4.8.2 against Remote Operations Console, operator console visualisations. Chartkit obliges the company to publish the modified files it distributes. BSD-3-Clause governs chartkit, which the inventory marks as no copyleft. chartkit was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The inventory entry OSS-010 is sqlite-embedded 3.45.1, on-robot local state. Sqlite-embedded was assessed for support as well as licence risk, and it carries a single-maintainer dependency with no succession plan. Edge Gateway Agent is where sqlite-embedded is consumed, which determines whether distribution occurs at all. sqlite-embedded is treated as material to the investment case rather than as a routine engineering item.

Jwt-verify carries MIT terms, classified no for copyleft purposes. Jwt-verify affects Identity and Authorization; no other component reviewed depends on it. Jwt-verify shows an approval status of approved 2024-06-21 against the company's own policy. jwt-verify is one of the entries where TDR-2026-19 relies on the company's own measurement rather than an independent one.

Csv-export sits inside Billing Usage Meter and is therefore in the shipped product rather than in tooling. OSS-026 covers csv-export at version 2.0.3, used billing and usage exports. Provenance of csv-export is public registry, signature verified. csv-export is the clearest instance in TDR-2026-19 of engineering practice being better than the documentation suggests.

The licence on image-signing is Apache-2.0; copyleft exposure is no. Image-signing shows an approval status of approved 2025-01-15 against the company's own policy. OSS-029 covers image-signing at version 2.5.0, used artefact signature verification. image-signing is the clearest instance in TDR-2026-19 of a control that exists on paper and is weaker in practice.

Linking mode matters for opencv-subset, which is used image preprocessing for grasp proposals. Opencv-subset carries a practical obligation: to attribute the component in the customer-facing notice file. Opencv-subset also carries a fork that must be maintained locally for as long as it is used, which is a supportability question rather than a licence one. opencv-subset carries a follow-up request that was satisfied after the draft of TDR-2026-19 was circulated.

Physics-core-fork carries this qualification: the fork diverges by nineteen patches and no upstream merge path is maintained. Licensing for physics-core-fork is Zlib and the copyleft column reads no. To share alike any derived asset it distributes is what physics-core-fork requires in practice. physics-core-fork is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Provenance of http-router is public registry, signature verified. The distribution question for http-router turns on the fact that it is used cloud service routing. In the approval log, http-router is approved 2024-01-09. http-router is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Telemetry Ingest is where tsdb-client is consumed, which determines whether distribution occurs at all. Tsdb-client is recorded as approved 2024-09-11 in the component approval log. No copyleft applies to tsdb-client, whose licence is MIT. tsdb-client was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Warehouse-sim-assets shows an approval status of approved with conditions 2024-11-06 against the company's own policy. Compliance with the warehouse-sim-assets terms requires to hold a current commercial licence for each developer seat. Warehouse-sim-assets reaches the customer only through Simulation Harness, and the reviewer confirmed there is no second path. warehouse-sim-assets was reviewed against the position twelve months earlier, and nothing measurable has changed.

OSS-001 records nav-graph-solver 2.4.1 against Motion Planner, linked into the planner as a static library. Where nav-graph-solver came from matters here: public registry, signature verified. Beyond licensing, nav-graph-solver carries a fork that must be maintained locally for as long as it is used. nav-graph-solver is the entry that changed most between the first and second document requests behind TDR-2026-19.

6. Security incidents and the follow-through on them

The reviewer checked the OTA Updater incident INC-2025-084 commitment against the code and release records; landed on 2025-07-14. Blue Mesa Components incident INC-2025-084 belongs to the state persistence group and sits with OTA Updater. Evidence strength on the OTA Updater incident INC-2025-084 analysis is moderate; the narrative is consistent but the reviewer could not independently reproduce it. The trigger recorded for state persistence incident INC-2025-084 on 2025-07-08 was that a paused update ring resumed after a routine service restart and eleven robots took an unintended firmware step. Following state persistence event INC-2025-084 the team undertook to persist the pause state and require a design note for any change to the release state machine, with Erik Palmgren accountable by 2025-08-15. state persistence incident INC-2025-084 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Taken with the rest of the record, state persistence incident INC-2025-084 indicates that the team's analysis was honest about a gap it could have concealed. The reviewer treats one contributing factor in OTA Updater incident INC-2025-084 as the more important lesson: the pause control had been added in a hotfix without a design review. The affected population in INC-2025-084 was one site and 11 robots. Reported detection applies to Blue Mesa Components incident INC-2025-084, since the site team escalated when robots rebooted outside the maintenance window. The post-mortem for state persistence incident INC-2025-084 identifies the cause as follows: the pause flag was held in process memory rather than in the release state record. state persistence incident INC-2025-084 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Incident INC-2025-096 was caused by a design gap: the routing table fell back to a mailbox rather than to a secondary responder. INC-2025-096 carries limited; the log retention window had closed before the reviewer asked for the raw data evidence, and the reviewer declined to raise the finding beyond what that supports. Wanda Njoku committed after incident INC-2025-096 to replace the mailbox fallback with a secondary responder and import rotations from the scheduling system, due 2025-10-15. Internal felt internal incident INC-2025-096, which affected no production site and no robot directly, the exposure being to internal systems. What incident INC-2025-096 tells the reviewer is that the fix is

real and the regression protection is not, so recurrence remains possible. process gap event INC-2025-096 is reported at the evidence strength available and has not been promoted to fit the narrative.

Internal incident INC-2025-096 began on 2025-08-05 when an overnight alarm routed to a shared mailbox because the on-call rotation had a two-hour gap. In taxonomy terms process gap incident INC-2025-096 is a process gap incident in Alarm Router. Alarm Router incident INC-2025-096 shows a short mitigation and a long tail: not applicable; no service impact against 26 days. For Alarm Router incident INC-2025-096 the follow-through was tested rather than assumed, and the fallback change landed on 2025-08-31; the rotation import was deferred to 2026. A contributing factor in process gap event INC-2025-096 was that the rotation source of truth was a spreadsheet maintained outside the alarm router. process gap incident INC-2025-096 is the clearest instance in TDR-2026-19 of a control that exists on paper and is weaker in practice.

Performance incident INC-2026-004 traces back to a single fact: the fallback scan matcher is slower than the primary and is selected whenever the map cache misses. Performance incident INC-2026-004 is recorded as follows on delivery: not landed; the matcher replacement is in design and the re-survey is scheduled. So far as reach is concerned, performance incident INC-2026-004 involved one site and 18 robots. For incident INC-2026-004 the reviewer holds moderate, resting on the team's own post-mortem and one supporting log extract evidence and reports the finding at that strength. Reading performance event INC-2026-004 from its first entry, on 2026-01-13 relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. performance event INC-2026-004 is one of the entries where TDR-2026-19 relies on the company's own measurement rather than an independent one.

The classification applied to performance incident INC-2026-004 is performance, and the affected component is Localization Engine. The two durations recorded for performance incident INC-2026-004 are not a service outage; 5 days to a workaround to mitigate and open at the review date to resolve. What performance incident INC-2026-004 tells the reviewer is that the incident is unremarkable in itself and revealing about the observability of the surrounding system. Attention came to INC-2026-004 when the site's commissioning engineer raised it during acceptance testing. The contributing factor recorded against Kestrel Medical Supply incident INC-2026-004 is that the site was surveyed with a sparse feature set that increases cache misses. INC-2026-004 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The engineering team's root cause for capacity incident INC-2025-139 is that control and telemetry traffic shared one topic and one consumer group budget, and the reviewer found the analysis sound. Incident INC-2025-139 began on 2025-11-02 when a telemetry backlog on a shared topic delayed control events by up to eleven seconds. The blast radius of multiple sites incident INC-2025-139 was 7 sites and 246 robots. In taxonomy terms INC-2025-139 is a capacity incident in Event Bus. Mitigation of capacity event INC-2025-139 took 2 h 47 m and full resolution took 63 days. INC-2025-139 was one of two items where TDR-2026-19 could not obtain the underlying data within the engagement.

Alerted detection applies to capacity event INC-2025-139, since the traffic manager's own latency alert fired first. The corrective action for multiple sites incident INC-2025-139 is to split control traffic onto a dedicated topic with its own quota and revisit the topic inventory quarterly, and 2026-01-31 is the date the team gave. Taken with the rest of the record, incident INC-2025-139 indicates that the failure was in the detection design rather than in the code that failed. Multiple sites incident INC-2025-139 would have been shorter or smaller had it not been that the topic inventory had recorded the sharing as acceptable in 2023 and it was never revisited. Verification of the capacity event INC-2025-139 remediation gives the split landed on 2026-01-04; the quarterly review has not yet run. capacity incident INC-2025-139 carries a follow-up request that was satisfied after the draft of TDR-2026-19 was circulated.

Time to mitigate Crestline Fulfillment incident INC-2025-041 was 1 h 52 m, and time to resolve was 9 days. So far as reach is concerned, resource leak event INC-2025-041 involved 2 sites and 61 robots. Traffic Manager incident INC-2025-041 rests on moderate; the narrative is consistent but the reviewer could not independently reproduce it evidence, which the reviewer states separately from the severity of the finding. For resource leak event INC-2025-041 the follow-through was tested rather than assumed, and landed on 2025-05-22, verified in the reservation replay suite. INC-2025-041 classifies as a resource leak event against Traffic Manager. resource leak event INC-2025-041 is one of the entries where TDR-2026-19 relies on the company's own measurement rather than an independent one.

Remediation for INC-2025-041 is to give every corridor reservation a lease with an expiry and add a site-level queue depth alert; Owen Brantley owns it and the committed date is 2025-05-30. Contributing to resource leak event

INC-2025-041, and not repaired by the remediation, is that the alert rule for queue depth existed but was scoped to a metric the site did not emit. 2025-03-14 is the date of resource leak event INC-2025-041, on which a robot lost its network link mid-traverse and its corridor reservation was never released. INC-2025-041 was caused by a design gap: the reservation lease had no expiry, so release depended on the departing robot. Resource leak event INC-2025-041 was detected because a site supervisor telephoned the on-call engineer after robots queued at one intersection. Traffic Manager incident INC-2025-041 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Reading stale state event INC-2025-078 from its first entry, on 2025-06-24 the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. Stale state event INC-2025-078 traces back to a single fact: the session list cached the site roster and only invalidated it on a page reload. INC-2025-078 was detected because an internal engineer noticed the discrepancy during a demonstration. Scope of impact in Harborline Systems incident INC-2025-078 was 4 site or tenant with no robot taken out of service, taken from the fleet records rather than from the narrative. Duration for INC-2025-078 splits into 1 h 15 m of impact and 34 days to close the underlying cause. stale state event INC-2025-078 rests on a single source, and the reviewer has said so rather than implying corroboration.

The judgement recorded against stale state event INC-2025-078 is that the failure was in the detection design rather than in the code that failed. Contributing to stale state incident INC-2025-078, and not repaired by the remediation, is that no test covered a roster change during an open operator session. Anjali Sekhon committed after stale state event INC-2025-078 to invalidate the roster cache on a fleet membership event and add a session-level regression test, due 2025-08-31. Moderate; the narrative is consistent but the reviewer could not independently reproduce it is the evidence grade for stale state event INC-2025-078, and the reader should not read severity into it. The reviewer checked the stale state incident INC-2025-078 commitment against the code and release records; landed on 2025-07-28; the regression test was still marked flaky at review. Harborline Systems incident INC-2025-078 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2026-19)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
give every corridor reservation a lease with an expiry and add a site-level queue depth alert	INC-2025-041	Owen Brantley	2025-05-30	landed	runbook change inspected
re-derive the time offset on every reconnect and add the affected variant to the second bench ri	INC-2025-047	Yuki Harada	2025-06-15	partly landed	no evidence located
release charge windows on a shift-end event and reconcile the calendar nightly	INC-2025-052	Nadia Semenova	2025-06-30	landed	configuration export inspected
key suppression on the tenant and identifier pair and backfill the affected order records	INC-2025-058	Tomas Villareal	2025-07-15	landed	runbook change inspected
make the allocator consult charger availability before excluding a robot	INC-2025-063	Marta Kellerman	2025-07-31	landed	interview only
add a leadership and under-replicated partition check to the broker upgrade runbook	INC-2025-071	Erik Palmgren	2025-07-31	landed	release record inspected
invalidate the roster cache on a fleet membership event and add a session-level regression test	INC-2025-078	Anjali Sekhon	2025-08-31	landed	interview only
persist the pause state	INC-2025-084	Erik Palmgren	2025-08-15	landed	commit inspected

Commitment	Source	Owner	Committed date	Status	Evidence inspected
and require a design note for any change to the release state machine					
add the connector torque check to incoming inspection and retrofit the affected harness batch	INC-2025-090	Bettina Vogel	2025-09-30	partly landed	runbook change inspected
replace the mailbox fallback with a secondary responder and import rotations from the scheduling	INC-2025-096	Wanda Njoku	2025-10-15	partly landed	runbook change inspected
add a deterministic tie-break and return the pinch-point scenario to the nightly suite	INC-2025-103	Priya Raghunathan	2025-10-31	partly landed	no evidence located
fail the load on an incomplete partition and reprocess the affected days	INC-2025-110	Farida Cisse	2025-10-31	landed	commit inspected
redact authorization headers at the logging layer and cut build log retention to fourteen days	INC-2025-118	Sonia Krall	2025-10-31	partly landed	interview only
annotate staleness on every read path and add a rebalance-under-load test to the platform suite	INC-2025-124	Marta Kellerman	2025-11-30	landed	runbook change inspected
restrict export to published revisions and shorten the publication path so the workaround is unn	INC-2025-131	Lucia Ferrante	2025-12-15	partly landed	test added and observed
split control traffic onto a dedicated topic with its own quota and revisit the topic inventory	INC-2025-139	Erik Palmgren	2026-01-31	partly landed	runbook change inspected
fold the reset into the re-run command and make double-counting detectable in the ledger	INC-2025-146	Nadia Semenova	2025-12-31	landed	no evidence located
apply the lifecycle rule through the configuration baseline and delete the over-retained clips	INC-2025-152	Anjali Sekhon	2026-01-31	partly landed	configuration export inspected

6. Security incidents and the follow-through on them - part 2 of 2

Set out next is the balance of the 6. Security incidents and the follow-through on them, in the sequence in which the rows were originally compiled. The break is editorial.

Everything stated about how the 6. Security incidents and the follow-through on them was assembled continues to hold, and the break carries no substantive meaning. Pagination alone determines where it falls.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2026-19) (continued, part 2 of 2)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
replace the fallback matcher and re-survey the affected	INC-2026-004	Rahul Menon	2026-06-30	not landed	commit inspected

Commitment	Source	Owner	Committed date	Status	Evidence inspected
aisles					
require a calibration record before returning a manipulator robot to service	INC-2026-009	Ines Duarte	2026-04-15	landed	interview only
move precondition validation to apply time and add an offline apply case to the platform suite	INC-2026-015	Sam Oyelaran	2026-03-31	landed	test added and observed
pin the full build configuration and add a determinism check to the harness	INC-2026-021	Colin Achebe	2026-05-31	not landed	release record inspected
group on the signal and block closure while a constituent alert remains active	INC-2026-028	Wanda Njoku	2026-04-30	landed	commit inspected
measure closing rate where the sensing permits and re-run the protective distance calculation	INC-2026-033	Bettina Vogel	2026-08-31	not landed	configuration export inspected
serialise cancellation against assignment and add an hourly orphan sweep	INC-2026-039	Grace Lindqvist	2026-05-31	landed	release record inspected
aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current h	INC-2026-045	Sam Oyelaran	2026-06-30	partly landed	interview only
check every template against the baseline and re-converge the four forked templates	INC-2026-051	Marcus Duplessis	2026-06-15	partly landed	release record inspected
refit the estimator on opportunity-charge data and add an imbalance alert independent of state o	INC-2026-057	Peter Halvorsen	2026-09-30	not landed	release record inspected
require a justification and a second approver within one working day for every emergency release	INC-2026-062	Colin Achebe	2026-08-31	not landed	release record inspected
add a slip detector that flags the condition and issue a site advisory on cleaning windows	INC-2026-068	Yuki Harada	2026-09-15	not landed	release record inspected
Partition the fleet allocator write path	TD-01	engineering leadership	2027-03-31	in design	register entry only
Replace the unmaintained scan-matching library in the localization fallback	TD-02	engineering leadership	2026-12-31	committed	register entry only
Retire the vendored geometry library in costmap fusion	TD-03	engineering leadership	2027-06-30	backlog	register entry only
Raise robot software line coverage above sixty per cent on the perception path	TD-04	engineering leadership	2027-03-31	in progress	register entry only
Bring the second bench rig to the current sensor configuration	TD-05	engineering leadership	2026-09-30	committed	register entry only

Commitment	Source	Owner	Committed date	Status	Evidence inspected
Make the nightly simulation suite blocking	TD-06	engineering leadership	2026-11-30	blocked	register entry only

7. Security debt and the funded remediation plan

About safety item TD-19 it should be said that the most significant open safety item in the reviewed record. TD-19 on Speed and Separation Monitor matters because it the protective distance rests on a fixed approach speed assumption. The register shows safety item TD-19 as in progress, due 2026-08-31. safety item TD-19 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Whether quality item TD-06 happens depends on funding, and it is not funded, and carried in the register regardless. Behind a supplier release outside the company's control is where register item TD-06 sits in the programme. TD-06 is the item to make the nightly simulation suite blocking, classified under quality. TD-06 on Simulation Harness was checked against the company's own register, which had not been updated since the change landed.

TD-18 on Safety Controller is sequenced behind a platform change that has its own target date. The plan treats safety evidence item TD-18 as funded only if the team's headcount plan is met, which the reviewer checked against the headcount model. If safety evidence item TD-18 is not done, the exposure is that it investigation evidence is lost between weekly extractions. programme item TD-18 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Sequencing of TD-08 on Remote Operations Console matters: it is last in a three-item chain, so its date is the least reliable on the register. The register entry programme item TD-08 reads: replace the archived charting library in the operator console, in the security supportability category. 1 engineer-quarter is the estimate on programme item TD-08, which converts to $132,000. security supportability item TD-08 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Configuration item TD-11 matters because it template drift produced a go-live failure in May 2026. The register entry Site Provisioning Service item TD-11 reads: re-converge the four forked site provisioning templates, in the configuration category. Site Provisioning Service item TD-11 is sized at 0.5 engineer-quarters, or $61,000 at the company's loaded rate. TD-11 illustrates why TDR-2026-19 separates the severity of a finding from the confidence behind it.

The quality entry at Payload Scale Interface item TD-22 is to move the calibration record for the payload scale to the cloud. Whether TD-22 happens depends on funding, and it is funded only if the team's headcount plan is met. Ahead of two other items that depend on it is where TD-22 sits in the programme. quality item TD-22 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-19.

Counsel was engaged after the reviewer raised it is the point the reviewer would put to management on TD-24. Costmap Fusion item TD-24 is funded in the current financial year in the engineering plan the reviewer was shown. $28,000 is the costed estimate for Costmap Fusion item TD-24, based on 0.25 engineer-quarters. Costmap Fusion item TD-24 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

TD-14 on Hardware-in-the-Loop Rig carries a 2027-03-31 target and a status of backlog. Register item TD-14 is the item to introduce a second bench rig queue to cut firmware lead time, classified under delivery. TD-14 carries this observation: requires capital for a third rig rather than engineering effort. Hardware-in-the-Loop Rig item TD-14 is reported at the evidence strength available and has not been promoted to fit the narrative.

8. Findings management accepted or declined to fix

For assessment finding SEC-2026-20 the reviewer notes that the underlying pattern appears in two other components that the tester did not reach. Payload scale calibration record can be edited without an audit entry is the substance of finding SEC-2026-20, raised on 2026-04-24. As at the review date finding SEC-2026-20 was accepted and the closure entry reads not planned. Finding SEC-2026-20 is rated Low with a numeric band of 3.2 on the tester's scoring scale. assessment finding SEC-2026-20 is one of the entries where TDR-2026-19 relies on the company's own measurement rather than an independent one.

Residual risk after remediation of SEC-2025-10 is low. Strong evidence supports Low-severity finding SEC-2025-10, namely that endpoint reachable and content captured. Remediated is the status of SEC-2025-10 against Charge Scheduler, dated 2025-10-09. Read alongside the remediation record, Charge Scheduler finding SEC-2025-10 shows that the risk acceptance lacks an expiry date, which is a process finding in its own right. assessment finding SEC-2025-10 is treated as material to the investment case rather than as a routine engineering item.

At 7.6 the score on Robot Shadow Store finding SEC-2026-02 places it in the High group. On assessment finding SEC-2026-02 the tester's evidence was that the tester read another tenant's shadow state from a test workload, and the reviewer grades it moderate. SEC-2026-02 against Robot Shadow Store records that robot shadow store cache lacked authorization on the internal read path, discovered on 2026-04-15. Read alongside the remediation record, High-severity finding SEC-2026-02 shows that the finding was closed within the committed window and the closure evidence was inspected. Robot Shadow Store finding SEC-2026-02 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Simulation harness cluster used a long-lived cloud role with broad read access is the substance of finding SEC-2026-06, raised on 2026-04-17. For SEC-2026-06 the record shows that role policy read from the cloud configuration export, which is strong evidence and is graded separately from severity. SEC-2026-06 translates into the following exposure: a compromised scenario could read production artefact storage. Assessment finding SEC-2026-06 produces this judgement: the fix is verified and the class of defect is not, since no test now prevents its return. SEC-2026-06 against Simulation Harness is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

What assessment finding SEC-2026-09 enables is this: a single query can exhaust the service's memory budget. Reading Fleet Health Monitor finding SEC-2026-09, the defect is that fleet health monitor query interface allowed unbounded result sets; the discovery date is 2026-04-18. Rating for Low-severity finding SEC-2026-09 is Low, band 3.9; the reviewer re-read the scoring rationale and accepted it. Remediated is the status of finding SEC-2026-09, dated 2026-05-08. finding SEC-2026-09 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

The residual position on finding SEC-2025-01 is low once lifetimes are cut to one hour with refresh, stated after mitigation rather than assuming full success. Impact of assessment finding SEC-2025-01 is not theoretical; a captured token remains usable long after the session that produced it. Reading assessment finding SEC-2025-01, the defect is that service tokens are issued with a twelve-hour lifetime; the discovery date is 2025-09-16. At 7.4 the score on Identity and Authorization finding SEC-2025-01 places it in the High group. SEC-2025-01 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

SEC-2026-10 against Billing Usage Meter is rated Low with a numeric band of 4.0 on the tester's scoring scale. Low-severity finding SEC-2026-10 is assessed as follows: the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. Impact of assessment finding SEC-2026-10 is not theoretical; an export could be altered after the fact. After the fix, finding SEC-2026-10 carries residual risk described as low; object lock is applied and verified nightly. finding SEC-2026-10 would be a candidate for the first post-close engineering review under any reasonable plan.

Remediation status on assessment finding SEC-2025-12 is accepted; the record shows not planned. The tester scored SEC-2025-12 against Analytics Warehouse Loader at 4.2 and labelled it Low. SEC-2025-12 sits against Analytics Warehouse Loader and records that analytics warehouse service account held write access to raw telemetry. Impact of SEC-2025-12 is not theoretical; a loader defect or compromise could alter source telemetry. Low-severity finding SEC-2025-12 was raised again with the owning engineer after the initial interview, and the position did not change.

9. Where a security failure would become a safety failure

Register entry HZ-19 was verified through occupancy sensor bench test plus a witnessed dock approach. What the register does not say about hazard HZ-19 in the register, and the reviewer does, is that occupancy sensing is fitted at every dock revision currently deployed. On history, Serious injury hazard HZ-19 produced no event, and the reviewer regards the absence as weak evidence given the exposure period. HZ-19 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-19.

After mitigation, hazard HZ-14 is Tolerable once the stale roster defect is closed; the register status is verified. The record shows that Occasional-exposure hazard HZ-14 produced two near-miss reports from site staff. The scope of hazard HZ-14 in the register is the sites that have moved to opportunity charging. hazard HZ-14 in the register is the reason one section of TDR-2026-19 concludes narrowly rather than generally.

Remote-exposure hazard HZ-21 is mitigated by push detection that engages the brake and requires a deliberate release. After mitigation, HZ-21 is Broadly acceptable; the register status is verified. Minor injury hazard HZ-21 was verified through witnessed push test on two robot variants. HZ-21 is the clearest instance in TDR-2026-19 of a control that exists on paper and is weaker in practice.

Obstacle detection bench test plus a site procedure review is what verified hazard HZ-13 in the register. Has produced no event, though the control has activated repeatedly is the incident history behind Occasional-exposure hazard HZ-13. Hazard HZ-13 is not clean: two sites use barriers below the guidance height and detection is then marginal. Occasional-exposure hazard HZ-13 was excluded from an earlier draft of TDR-2026-19 and reinstated after the reviewer obtained the supporting record.

Occasional-exposure hazard HZ-02 covers the case where robot contacts a pedestrian emerging from a blind cross-aisle. Against Occasional-exposure hazard HZ-02 the team relies on aisle-end speed reduction plus a mirrored sensing field at mapped blind corners. The reviewer's note on hazard HZ-02 in the register is that the reviewer confirmed mapping at four of six sites and could not evidence the remaining two. HZ-02 is treated as material to the investment case rather than as a routine engineering item.

For hazard HZ-15 the verifying activity was site commissioning check plus a fire drill observation at two sites. Mitigation for hazard HZ-15 is egress routes marked as keep-out and a stop-and-clear behaviour on a fire signal. On history, register entry HZ-15 has produced no event, though the control has activated repeatedly. Serious injury hazard HZ-15 was cross-checked against the engineering record supplied for TDR-2026-19, and the two agree.

For hazard HZ-16 the verifying activity was two rollback drills plus a fleet version audit. HZ-16 identifies the hazard that firmware update leaves a robot in a mixed-version state. HZ-16 carries residual risk of Tolerable with a status of verified. Remote-exposure hazard HZ-16 is one of the entries where TDR-2026-19 relies on the company's own measurement rather than an independent one.

Serious injury hazard HZ-09 scores Serious injury on severity; exposure is Remote and controllability Medium. HZ-09 has produced no event, though the control has activated repeatedly; that is worth stating alongside the assessed exposure. Where Remote-exposure hazard HZ-09 is concerned, the hazard is that robot fails to stop on a descending ramp when a wheel loses traction. HZ-09 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Minor injury hazard HZ-12 applies to every robot variant currently in production. Hazard HZ-12 in the register carries residual risk of Tolerable with a status of verified. In the reviewed period hazard HZ-12 produced two near-miss reports from site staff. Occasional-exposure hazard HZ-12 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

10. Open findings, risk acceptances and qualifications

Where the report at TDR-2026-19 cannot conclude, it says what evidence would let it conclude. L. Casaubon has dated each open item in the report at TDR-2026-19 so that a reader can see which have been open longest rather than only how many there are. Pinehurst Systems Assurance delivered this report on 2026-05-08; the scope agreed with Ardent Peak was the security architecture, penetration-test findings and vulnerability management of the platform. Scope of this report is the security architecture, penetration-test findings and vulnerability management of the platform, and the cut-off for evidence is 2026-05-08. The security architecture, penetration-test findings and vulnerability management of the platform was examined over 7 weeks, which sets a real limit on the depth reached in the later sections.

Where management has declined to act, the present report preserves the decision and the stated rationale so a post-close plan can revisit it. 4 declined items are carried in the present report, of which two lack any documented compensating control. Pinehurst Systems Assurance notes that 1 of the matters in this report are capable of being closed before completion if the company is asked in time. L. Casaubon would put 6 items from this report into the first hundred

days rather than into a warranty. Pinehurst Systems Assurance has not been asked to opine on valuation, and nothing in this report should be read as doing so.

L. Casaubon treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and this report keeps the two apart. The convention throughout the report at TDR-2026-19 is that an absence of evidence is reported as such rather than as evidence of absence. The present report does not include a code audit; the reviewer read code where a specific claim required it and no further. The reviewer notes that the report at TDR-2026-19 was prepared while three remediation programmes were in flight, so several positions will have moved by completion. Pinehurst Systems Assurance has flagged 1 matters for the transaction record and left the remainder as engineering work.

The report at TDR-2026-19 would have the monitoring pack include declined items with their compensating controls, so that acceptance decisions age visibly. L. Casaubon would add one qualitative item to the pack proposed in the present report: the count of commitments closed without verifying evidence. This report addresses the security architecture, penetration-test findings and vulnerability management of the platform. It does not address commercial terms, financial reporting or employment matters, which sit with other workstreams. Pinehurst Systems Assurance was instructed by Ardent Peak to examine the security architecture, penetration-test findings and vulnerability management of the platform; the resulting report is this report, dated 2026-05-08. Any figure in the present report that the reviewer could not trace to a system is attributed to the person who gave it.

11. Conclusions on the security position

TDR-2026-19 concludes that the technical debt register is honest about what exists and optimistic about when it will be addressed. On the evidence available by 2026-05-08, the report at TDR-2026-19 concludes that the technical debt register is honest about what exists and optimistic about when it will be addressed. Evidence gathering for the report at TDR-2026-19 was iterative: an initial document request, 13 interviews, then a second request that closed most of the gaps. The present report follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. The numbering in report TDR-2026-19 follows the order of the work rather than the order of importance.

None of the findings in the present report would in the reviewer's view justify withholding completion, and two would justify a specific undertaking. The costed remediation implied by the present report is $780,000 over the coming four quarters, which is within the ordinary engineering budget. Management's declinatures are reported in the report at TDR-2026-19 at face value; the reviewer's disagreement is stated separately and does not overwrite them. TDR-2026-19 treats a backlog label as evidence of intent and not as evidence of remediation, which is why several items remain open in this report that the company regards as handled. The present report does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

Monitoring recommended by report TDR-2026-19 relies on metrics the company already emits, so no new instrumentation is required to start. The present report proposes 5 metrics for the post-close monitoring pack, each already produced by an existing system. The present report keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. L. Casaubon asked in every declined case who would revisit the decision and when, and report TDR-2026-19 records the answer even where there was none. Report TDR-2026-19 was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.